Simple Linear Regression Model
ECO 3121 Tutorial

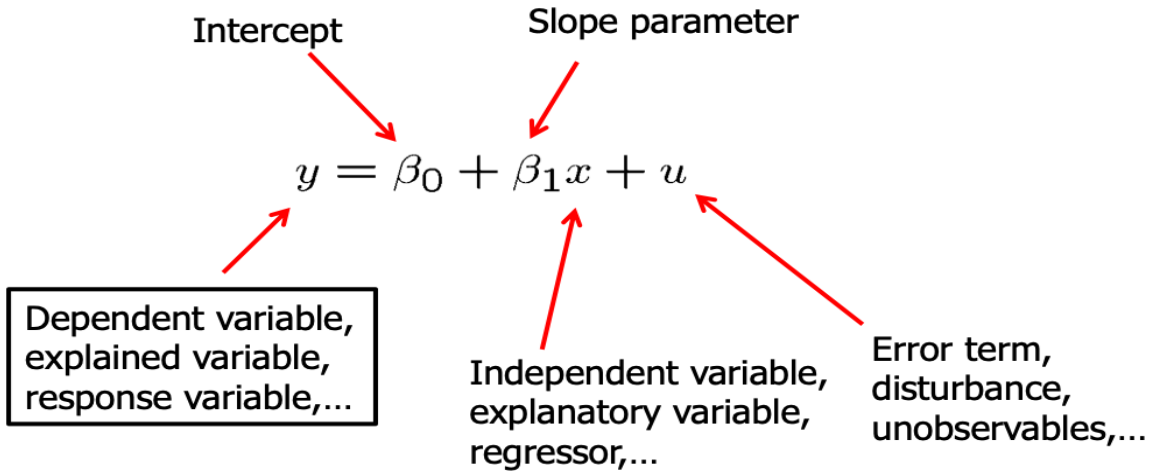
Week 3 tutorial

What is linear regression?

- Linear Regression is a statistical technique that uses one or several explanatory variables to predict the outcome of a dependent variable. We want to capture the variation of the dependent variable

"Explains variable y in terms of variable x "

For instance, x is amount of fertilizer, and y is soybean yield; x is years of schooling, y is hourly wage.



Why we want to use linear regression: (The Gauss-Markov Theorem)

- The Gauss-Markov Theorem: Under classical assumptions, then ordinary least squares (OLS) regression produces unbiased estimates that have the smallest variance of all possible linear estimators.

Assumption SLR.1 (Linear in Parameters)
In the population model, the dependent variable, y, is related to the independent variable, x, and the error (or disturbance), u, as

$$y = \beta_0 + \beta_1 x + u,$$

where β_0 and β_1 are the population intercept and slope parameters, respectively.

Assumption SLR.2 (Random Sampling)
We have a random sample of size n, $\{(x_i,y_i): i = 1, 2, \ldots, n\}$, following the population model in Assumption SLR.1.

Assumption SLR.3 (Sample Variation in the Explanatory Variable)
The sample outcomes on x, namely, $\{x_i, i = 1, \ldots, n\}$, are not all the same value.

Assumption SLR.4 (Zero Conditional Mean)
The error u has an expected value of zero given any value of the explanatory variable. In other words,

$$E(u|x) = 0.$$

Assumption SLR.5 (Homoskedasticity)
The error u has the same variance given any value of the explanatory variable. In other words,

$$Var(u|x) = \sigma^2.$$

How can we obtained the coefficient estimator

- **What does "as good as possible" mean?**
- Regression residuals

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

- Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize sum of squared regression residuals

$$\min \sum_{i=1}^{n} \hat{u}_i^2 \quad \rightarrow \min \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- First order conditions

$$n^{-1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$n^{-1} \sum_{i=1}^{n} x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

How should we interpret the coefficient estimator

Model	Dep. Var.	Indep. Var.	β_1
Level-Level	y	x	$\Delta y = \beta_1 \Delta x$
Level-Log	y	$\log(x)$	$\Delta y = (\beta_1/100)\%\Delta x$
Log-Level	$\log(y)$	x	$\%\Delta y = (100\beta_1)\Delta x$
Log-Log	$\log(y)$	$\log(x)$	$\%\Delta y = \beta_1 \%\Delta x$

Some Practical problems

1 Let *kids* denote the number of children ever born to a woman, and let *educ* denote years of education for the woman. A simple model relating fertility to years of education is

$$kids = \beta_0 + \beta_1 educ + u,$$

where u is the unobserved error.

(i) What kinds of factors are contained in u? Are these likely to be correlated with level of education?

(ii) Will a simple regression analysis uncover the ceteris paribus effect of education on fertility? Explain.

Some Practical problems

4 The data set BWGHT.RAW contains data on births to women in the United States. Two variables of interest are the dependent variable, infant birth weight in ounces (*bwght*), and an explanatory variable, average number of cigarettes the mother smoked per day during pregnancy (*cigs*). The following simple regression was estimated using data on $n = 1{,}388$ births:

$$\widehat{bwght} = 119.77 - 0.514\, cigs$$

(i) What is the predicted birth weight when $cigs = 0$? What about when $cigs = 20$ (one pack per day)? Comment on the difference.

(ii) Does this simple regression necessarily capture a causal relationship between the child's birth weight and the mother's smoking habits? Explain.

(iii) To predict a birth weight of 125 ounces, what would *cigs* have to be? Comment.

(iv) The proportion of women in the sample who do not smoke while pregnant is about .85. Does this help reconcile your finding from part (iii)?